

Arnav Pundir

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PROFESSIONAL SUMMARY

Computer Vision and AI-focused Software Engineer with hands-on expertise in Face Recognition, Object Detection, Temporal Tracking, and Natural Language Processing. Proven track record of engineering deployable real-world AI systems using YOLOv8, OpenCV, MediaPipe, and LLMs (Gemini). Highly skilled in integrating robust machine learning pipelines into scalable full-stack web applications using Python, FastAPI, and React.js to deliver high-impact automation and analytics.

TECHNICAL SKILLS

Languages : Python, C/C++, JavaScript, TypeScript, SQL
ML & AI : OpenCV, YOLOv8, MediaPipe, TensorFlow/Keras, InsightFace, NLP, LLM (Gemini API)
Backend & Web : FastAPI, Flask, React.js, Next.js, REST APIs, WebRTC
Tools & Cloud : Git, GitHub, Docker, VS Code, Azure AI Vision, MongoDB, PostgreSQL

EDUCATION

COER University	Roorkee, India
<i>Bachelor of Technology in Computer Science and Engineering</i>	2024 – 2028
Atal Utkrisht Gov. Inter College	Roorkee, India
<i>Senior Secondary (Class XII) PCM</i>	2022 – 2024

PROFESSIONAL EXPERIENCE

Freelance Software Developer	Jan 2024 – Present
<i>Client & Independent Projects</i>	Roorkee, India
- Engineered scalable, high-performance web applications (e.g., SevaSetu) using React.js and robust REST APIs, evaluating architectural trade-offs to choose optimal solutions for complex client requirements. - Architected interactive WebGL applications utilizing Three.js, critically analyzing system performance bottlenecks to consistently achieve 95+ Lighthouse accessibility and speed scores.	

AI & COMPUTER VISION PROJECTS

TraceAI Forensic Surveillance Intelligence Platform | [GitHub](#)

- Developed a real-time AI-powered forensic surveillance system applying advanced computer vision algorithms (**YOLOv8**) to accurately track and re-identify individuals across distributed video sources.
- Engineered a robust temporal tracking pipeline to process large-scale video feeds, successfully constructing precise movement timelines for complex visual identification challenges.

Face Recognition Attendance Web System | [GitHub](#)

- Automated attendance marking from uploaded group photos by deploying a highly optimized **InsightFace** encoding pipeline, achieving **90%+** recognition accuracy in multi-face detection.
- Integrated the deep learning pipeline within a **Flask** backend and built a seamless web interface, reducing manual attendance effort by **80%** with automated CSV report generation.

DocBot AI Documentation Assistant | [GitHub](#)

- Architected an intelligent NLP application leveraging the **Gemini API (LLM)** to autonomously summarize complex web documentation and answer natural language user queries in real-time.
- Built a highly responsive native GUI using **PyQt5**, efficiently orchestrating live content extraction, semantic text processing, and prompt engineering workflows.

Traffic Analysis System | [GitHub](#)

- Built a real-time vehicle detection and counting pipeline using **YOLOv8** and **OpenCV**, optimizing inference to process high-definition video streams at **25+ FPS**.
- Engineered multi-class detection algorithms to generate structured CSV-based traffic flow analytics for smart city applications and congestion monitoring.

CERTIFICATIONS & OPEN SOURCE

Nexus Spring of Code (NSoC '26) & GSSoC	Global Rank #63
<i>Core Contributor</i>	Apr 2026 – Present
Merged 13 Pull Requests across 5 repositories, actively analyzing software design to mitigate vulnerabilities and optimize performance for production environments.	
Microsoft Learn & Google	2025 – 2026
<i>Azure AI Vision, Introduction to GenAI, GitHub Copilot</i>	Online
- Earned Microsoft Learn trophies in Object Detection, OCR, and Face Recognition. - Certified in Prompt Engineering and foundational concepts of Large Language Models by Google and GitHub.	